

Generation Interconnection System Impact Study Report

For Queue Project Q17-0805

Jarrell East 100 kV

October 29, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0805. The Interconnection Customer proposes a 123.2 MW solar photovoltaic facility interconnecting at the Jarrell East 100 kV substation in the Illinois area. The request was received on June 05, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 322 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,131,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Swan Valley Substation to Eastown Rd 230 kV	Swan Valley to Trego 500 kV Line	104.1	614	6.3
Scott to Shawville Station 345 kV	Orchard Rd to Shawville Station 500 kV	103.6	1,626	5.1
Galion Generating Station 500/230kV TX	Trego to Beaver Substation 500 kV	100.3	1,059	10.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Swan Valley Substation to Eastown Rd 230 kV	Reconductor line	\$3,075,000	1% (\$37,000)
Scott to Shawville Station 345 kV	Reconductor line	\$42,580,000	2% (\$785,000)
Galion Generating Station 500/230kV TX	Replace transformer with higher-rated unit	\$28,322,000	1% (\$309,000)
	Total allocated to project		\$1,131,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.